DATABASE DESIGN

MongoDB

User Tables:

Field Name	Data Type	Description
user_id	INT (PK)	Unique user ID
full_name	VARCHAR(100)	User full name
username	VARCHAR(50)	Login username
password	VARCHAR(255)	Encrypted password
role	VARCHAR(20)	Admin / Librarian / Member
email	VARCHAR(100)	Email address
phone	VARCHAR(15)	Contact number
status	VARCHAR(10)	Active / Inactive
created_at	DATETIME	Account creation date

Book Tables:

Field Name	**Data Type**	**Description**
book_id	INT (PK)	Unique book ID
title	VARCHAR(150)	Book title
author	VARCHAR(100)	Author name

Field Name	Data Type	Description
publisher	VARCHAR(100)	Publisher
category	VARCHAR(50)	Book category
isbn	VARCHAR(20)	ISBN number
edition	VARCHAR(20)	Book edition
total_copies	INT	Total number of copies
available_copies	INT	Available copies
added_date	DATE	Book added date

Book Issue Tables:

Field Name	Data Type	Description
issue_id	INT (PK)	Unique issue ID
user_id	INT (FK)	User ID
book_id	INT (FK)	Book ID
issue_date	DATE	Date of issue
due_date	DATE	Return due date
return_date	DATE	Actual return date
status	VARCHAR(20)	Issued / Returned

Fine Tables:

Field Name	Data Type	Description
fine_id	INT (PK)	Unique fine ID
issue_id	INT (FK)	Issue ID
user_id	INT (FK)	User ID
days_late	INT	Number of late days
fine_amount	DECIMAL(8,2)	Fine amount
payment_status	VARCHAR(20)	Paid / Unpaid
paid_date	DATE	Date of payment